Crime Data Analysis Dashboard

Problem Statement

- Crime rates are **increasing continuously**
- Different cities have **different crime patterns**
- Large amount of data is available but **not properly analysed**
- Difficult for authorities to **take effective decisions**
- Need to convert raw data into **meaningful insights**

Project Overview

What This Project Covers

Dataset at a Glance

Analysis of a **crime dataset (500 records)**

- City, Crime Type, Arrest Status
- Gender, Severity, Weapon

Approach

Used **Excel Pivot Tables & Charts**

Focus on identifying **patterns & trends**

Converts data into **easy insights**

Project Objectives

What We Aim to Achieve

High Crime Cities

Identify **high crime cities**

Crime Types

Find **most common crime types**

Arrest Comparison

Compare **arrested vs non-arrested cases**

Gender Distribution

Analyse **gender distribution of victims**

Severity Levels

Study **severity levels of crimes**

Weapon Usage

Understand **weapon usage patterns**

Dataset Description

About the Data

Total Records

500

Rows

Each row represents **one crime case**

Columns Include

01

Crime_ID

02

City

03

Crime_Type

04

Arrest_Status

05

Victim_Gender

06

Severity_Level

07

Weapon_Used

08

Time

Data Cleaning & Preparation

Preparing the Data for Analysis

Removed Blank Values

Removed **blank values** to ensure data completeness

Time Format Conversion

Converted **time format (24 hr → 12 hr)** for readability

Correct Format

Ensured data is in **correct format** throughout

New Fields Created

Created new fields like **Day/Night column**

Ready for Analysis

Data prepared for **accurate analysis**

Tools & Techniques Used

How the Analysis Was Done

Primary Tool

Microsoft Excel

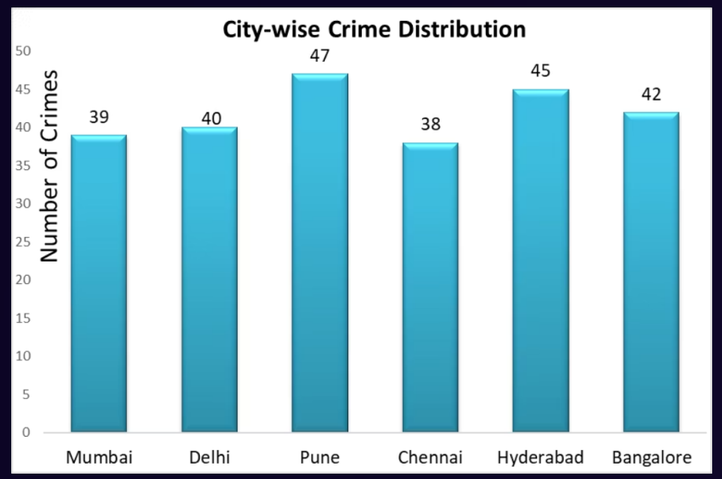
Pivot Tables for analysis

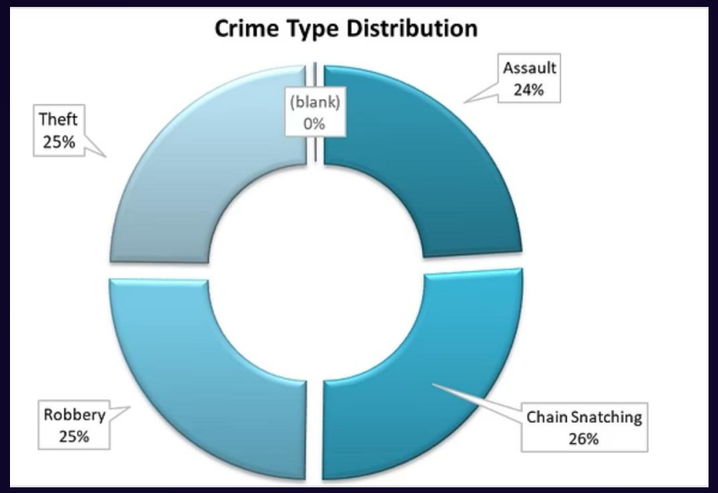
Slicers for filtering

Data visualisation techniques

Charts Used

- Bar Chart
- Pie Chart
- Column Chart

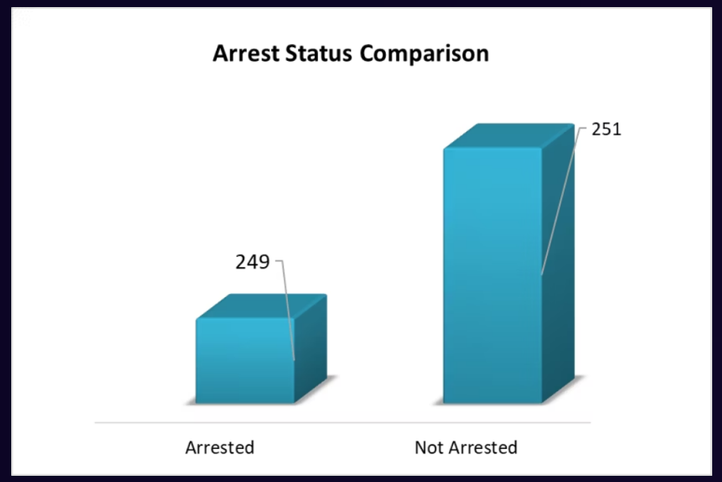

City-wise Crime Distribution
Number of Crimes
50
45
40
35
30
25
20
15
10
5
0
39
40
47
38
45
42
Mumbai
Delhi
Pune
Chennai
Hyderabad
Bangalore

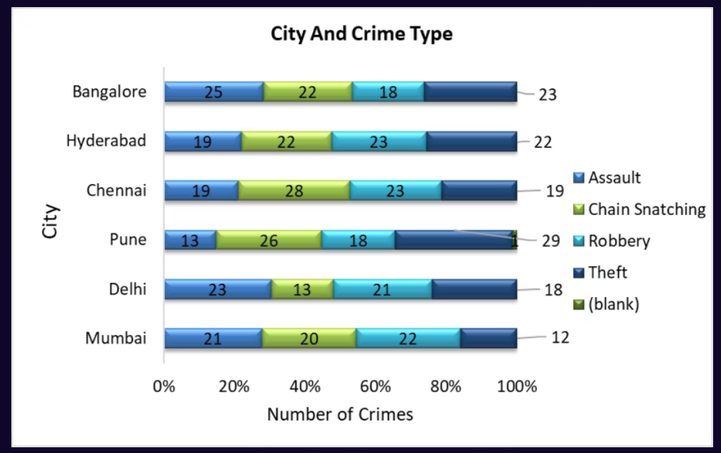

Crime Type Distribution
Assault
24%
(blank)
0%
Theft
25%
Robbery
25%
Chain Snatching
26%

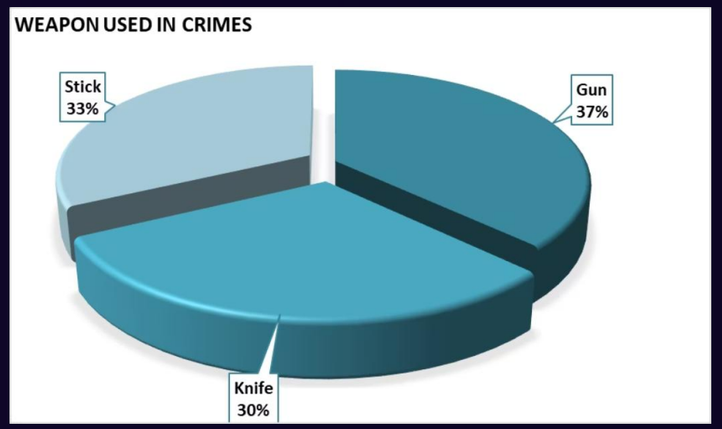
VICTIM GENDER DISTRIBUTION
Male
49%
Female
51%

chart used:

charts:


Arrest Status Comparison
249
251
Arrested
Not Arrested


City And Crime Type
Bangalore
Hyderabad
Chennai
Pune
Delhi
Mumbai
City
25 22 18 23
19 22 23 22
19 28 23 19
13 26 18 29
23 13 21 18
21 20 22 12
0% 20% 40% 60% 80% 100%
Number of Crimes
Assault
Chain Snatching
Robbery
Theft
(blank)


WEAPON USED IN CRIMES
Stick 33%
Gun 37%
Knife 30%

Who This Dashboard Is For?

Target Audience

Law Enforcement Agencies

Empowering officers with **crime analysis** insights

Government Authorities

Helping authorities make **data-driven decisions**

Data Analysts

A rich dataset for **crime analysis** exploration

Researchers

Anyone interested in **crime analysis**

Dashboard Overview

Interactive Crime Analysis Dashboard

Interactive dashboard using **slicers** — easy to **filter and analyse data**, helping in **quick decision-making**.

City-wise Crime

Breakdown of crimes across different cities

Crime Type Distribution

Visual breakdown of most common crime types

Arrest Analysis

Comparison of arrested vs non-arrested cases

Gender & Severity Insights

Gender distribution and severity level analysis